

Page 1

Carbon & its Compounds

Page 2

5

Page 3

5

Page 4

Carbon and its Compounds *Bonding in Carbon (covalent bond): Ionic compounds have high melting and boiling points and conduct electricity in solution or in the molten state. -Most carbon compounds are poor conductors of electricity. ■ Since these compounds are largely non-conductors of electricity, we can conclude that bonding in these compounds does not give rise to any ions. -The atomic number of Carbon is 6. - In case of carbon, it has 4 electrons in its outermost shell and needs to gain or lose 4 electrons to attain noble gas configuration. = If it were to gain or lose electrons → i) It could gain 4 electrons forming C⁴⁻ anion. -But it would become difficult for the nucleus to hold 10 electrons (4 extra gained). ii) It could lose 4 electrons forming C⁴⁺ cation. -But it would require a large amount of energy to remove 4 electrons leaving behind a carbon cation with 6 protons in its nucleus holding on to just 2 electrons.

Page 5

→To overcome this, Carbon shares its valence electrons (with carbon or other elements) to attain noble gas configuration. -Many other elements share their valence electrons to attain noble gas configuration. →Let us take an example : →Your friend Ashish had 2 pens. You did not have any pen. You asked him for a pen, he gave 1 pen to you and asked you to keep it. This way Ashish lost one pen but you gained one pen. This is same as losing and gaining of electrons. • But when you talk about SHARING, Ashish had 1 pen and you had no pen, so Ashish shared a pen with you where he is also using the pen and giving to you when you need it. This is same as sharing of electrons. * Electrons in the outermost shell are called VALENCE ELECTRONS. →The atomic number of Hydrogen is 1. It has 1 electron in its K shell (Atomic no-1). So, two Hydrogen atoms share their electrons to form a molecule of Hydrogen, H₂. Hence, attaining noble gas configuration. (1 electron Pair share, Single (-) bond) (2 electrons Pair share, double (=) bond) (3 electrons Pair share, triple (≡) bond) Some more examples : ① O → Atomic number → 8 Configuration → 2,6 Valence Electrons → 6

Page 6

②N → Atomic no. (Nitrogen) → 7 * Configuration → 2,5 * Valence Electrons → 5 Allotropes of Carbon → The element carbon occurs in different forms in nature with widely varying physical Properties. Graphite Graphite: →In graphite, each carbon atom is bonded to three other carbon atoms in the same plane giving a hexagonal array. Buckminster Fullerene Buckminster Fullerene C-60: 20 (six) membered rings 12 (five) membered rings Diamond Diamond: In diamond, each carbon atom is bonded to four other carbon atoms forming a rigid three-dimensional structure.

Page 14

Assertion Reasoning

Page 15

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